Transformer belief state geometry on a non-ergodic hidden Markov model

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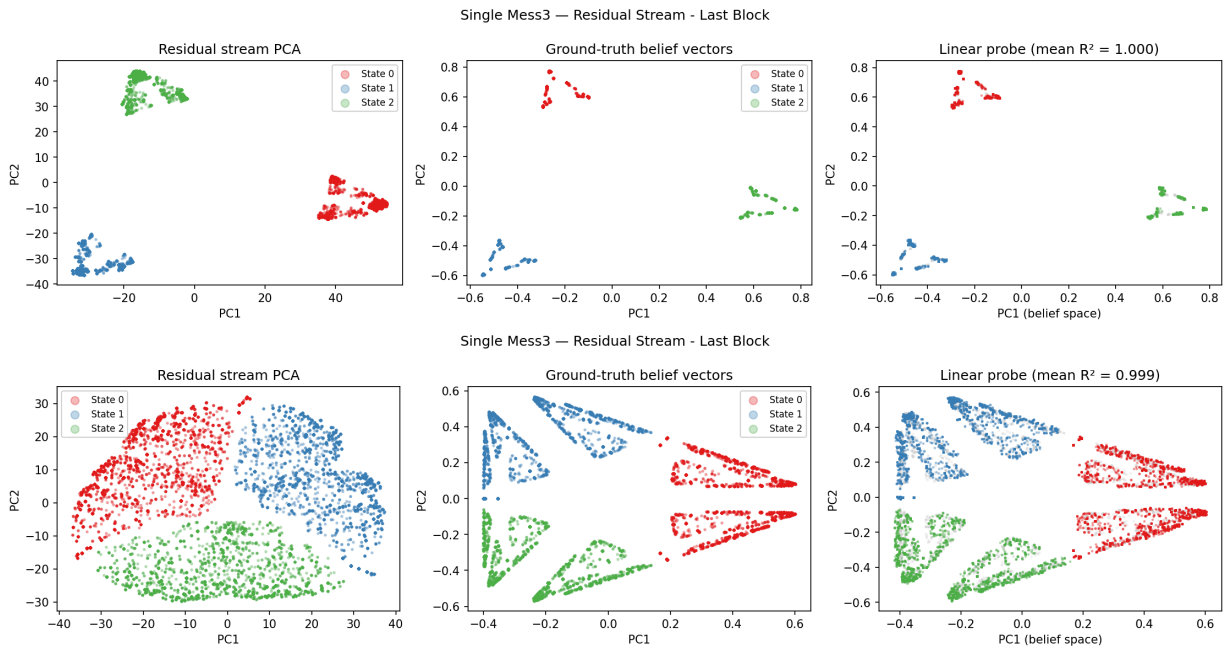
As a sanity check, I first trained a transformer on sequences from a single Mess3 process to verify that the network can represent belief state geometry in its residual stream. I tested two processes: Mess3($\alpha = 0.95, x = 0.1$) and Mess3($\alpha = 0.05, x = 0.1$). The first one has low temporal autocorrelation and a diffuse, slowly-evolving belief state, and the second one has high self-predictability and a rapidly-concentrating belief state. The transformer network has [*d_model = 120, d_mlp = 480, n_heads = 3, n_layers = 4, seq_len = 8*] and was trained with [*optimizer = Adam, lr = 5e-4, batch_size = 25000, steps = 5000*]. After training, I fit linear probes on the final-block residual stream following the method of Shai et al. (2024). In both cases, the probe decoded the belief state with $R^2 \approx 1.0$, confirming that the network learns to represent belief states linearly in its residual stream. I then used the same architecture and training hyperparameters for all subsequent experiments, with sequence length extended to 16.



I then trained the transformer network on a non-ergodic Mess3 mixture: component A is Mess3($\alpha = 0.95, x = 0.1$) and component B is Mess3($\alpha = 0.05, x = 0.1$). Each training sequence is generated by one of the two components with equal probability. This is different from the factored case studied in Shai et al., 2026, where multiple processes run in parallel within a single sequence. Here the latent component identity is fixed per sequence but unknown to the model.

This setup is interesting and relevant because it captures a senario that arises naturally for language models. Specifically, a model often needs to infer an unknown latent cause from accumulating context in order to make accurate predictions. For example, consider a setting where the generative process depends on some unknown property of the user (e.g., the identity of the user). The model has no access to this information a priori, but must infer it from the sequence itself. Correctly identifying the latent cause is critical, and a failure to do so can lead to systematic prediction errors. This makes the non-ergodic mixture a natural and tractable testbed for studying latent cause inference in transformers.

We can start from an intuitive analysis of the problem. In the single Mess3 case, the model must infer which of three hidden world states it currently occupies, so the relevant belief state is a distribution over world states: $\eta(t) \in \Delta^2$. In the non-ergodic mixture case, the model faces an additional inference problem: it must simultaneously infer the hidden world state and the latent component generating the current sequence. This is an example of latent state inference, a problem that is extensively studied in reinforcement learning. In symbolic models of this problem class such as the Bayes-Adaptive MDP (BAMDP), the standard solution is to augment the belief state by incorporating the latent cause into the state space. The agent then maintains and updates a joint posterior over world states and latent causes. Thus, we should expect the transformer to represent the posterior over component identity, $\mathrm{post}_A(t) = P(c = A \mid x_{0:t})$, where c is the latent component.

More concretely, we can examine what information is strictly necessary to compute the Bayes-optimal next-token prediction:

$$P(x_{t+1} \mid x_{0:t}) = \mathrm{post}_A(t) \cdot P_A(x_{t+1} \mid \eta_A(t)) + \mathrm{post}_B(t) \cdot P_B(x_{t+1} \mid \eta_B(t)),$$

where

$$\mathrm{post}_A(t) = \frac{\mathrm{post}_A(t-1) p_A(x_t \mid \eta_A(t-1))}{\mathrm{post}_A(t-1) p_A(x_t \mid \eta_A(t-1)) + \mathrm{post}_B(t-1) p_B(x_t \mid \eta_A(t-1))},$$

and similar for $\mathrm{post}_B(t)$. Prior work has shown that neural networks (specifically, memory-based meta-learning agents) trained on next-token prediction converge to near-Bayes-optimal solutions and learn to encode belief states in their hidden states consistent with the optimal predictor (Ortega et al. 2019; Genewin, 2025). To make Bayes-optimal prediction, the agent therefore needs to maintain three quantities: $\eta_A(t)$, $\eta_B(t)$, and $\mathrm{post}_A(t)$. Interestingly, both $\eta_A(t)$ and $\eta_B(t)$ must be represented simultaneously, regardless of which component generated the sequence. This is a non-trivial prediction: even for a sequence drawn from component A, the model must internally maintain and update the belief state it would have *as if the sequence came from component B*. In other words, the model is required to run two parallel belief update processes, one for each component, and weight their predictions by the inferred posterior $\mathrm{post}_A(t)$.

Based on the above, we can make the following predictions about the geometry of the residual stream. At any given context position t, the residual stream must encode the triple $(\eta_A(t), \eta_B(t), \mathrm{post}_A(t))$. Since η_A and η_B each live in a 2-simplex (Δ^2, two degrees of

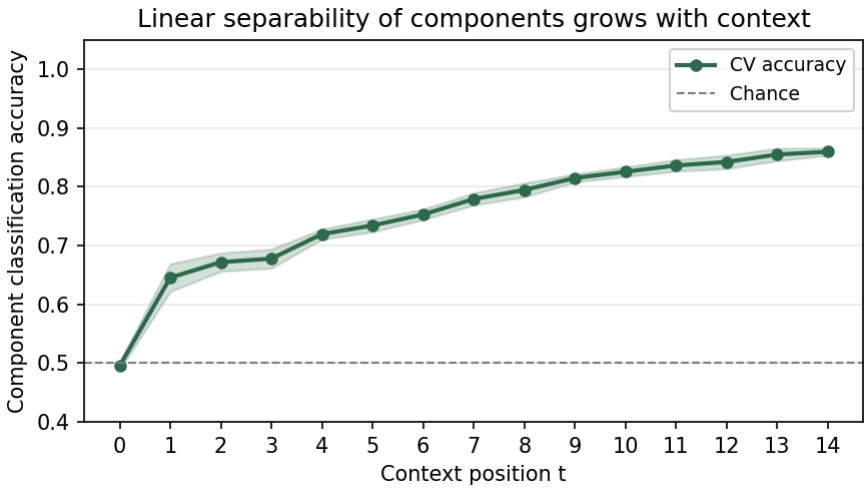
freedom), and post_A is a scalar, the minimal representational space required is five-dimensional. The simplest geometric prediction is that the residual stream represents these three quantities as approximately linear, and in approximately orthogonal subspaces. Since η_A and η_B are updated by different transition matrices, the two belief-state update rules are independent. Thus, interference between their representations is costly and orthogonality is the efficient solution.

It is worth noting that the theoretical justification for orthogonality in this setting seems to be more subtle than it may appear. The factored world hypothesis and Theorem 2.3 of Shai et al. (2026) guarantee orthogonal subspaces when the joint transition operator factorizes as a tensor product $T^{(x)} = T_A^{(x)} \otimes T_B^{(x)}$, meaning both processes run in parallel and update independently on every token. This condition holds in the factored setting of Shai et al. but does not seem to hold here. In the mixture setting, the generative process is not a tensor product of two GHMMs, but only one component is ever active per sequence. The model maintains η_A and η_B simultaneously not because the world is a product of two independent processes, but as a computational strategy for Bayesian inference under uncertainty about which process generated the sequence. The effective joint transition operator is a posterior-weighted combination $T^{(x)} = \mathrm{post}_A \cdot T_A^{(x)} + \mathrm{post}_B \cdot T_B^{(x)}$, which does not satisfy the definition in Shai et al. and therefore falls outside the scope of their theorem. The orthogonality we predict is therefore better understood as an efficiency argument: representing η_A and η_B in overlapping subspaces would cause interference between two belief streams that carry different information. The network has a functional incentive to separate them, and orthogonality is the maximally efficient solution. Whether this efficiency argument can be made rigorous in the mixture setting is an interesting open theoretical question.

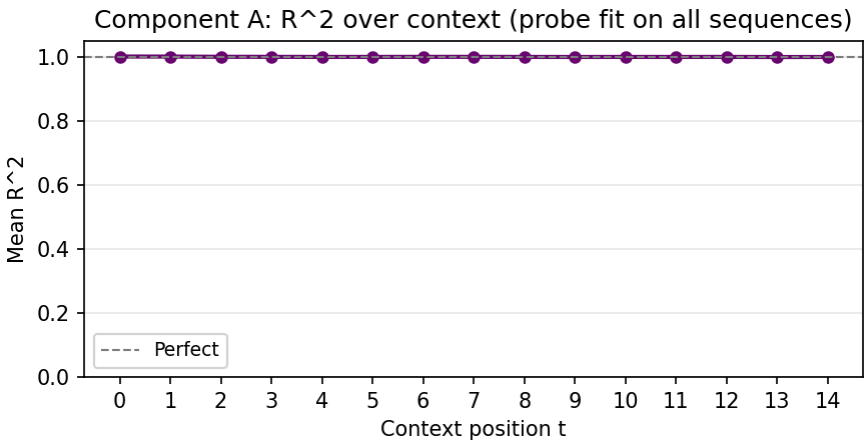
How should the geometry evolve with context position and layers? At $t = 0$, the model has observed no tokens, so $\eta_A(0) = \eta_B(0)$ and $\mathrm{post}_A = 0.5$. All sequences map to the same point, so the geometry collapses to a single point. As t increases, $\eta_A(t)$ and $\eta_B(t)$ begin to diverge as the belief states respond to the token sequence, spreading the activations across the two simplices. Simultaneously, $\mathrm{post}_A(t)$ moves away from 0.5 toward 0 or 1 as the component identity becomes clearer. By late context positions, sequences generated by component A will have $\mathrm{post}_A \approx 1$, while component B sequences will have $\mathrm{post}_A \approx 0$. I therefore predict that the effective dimensionality of the residual stream geometry increases with context position, as the initially collapsed point expands into a structured five-dimensional manifold. Across layers, I predict a progressive refinement of the mixture computation. Concretely, the belief state geometry should emerge across layers.

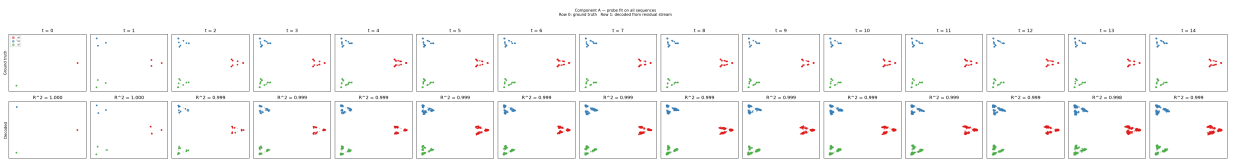
Alternatively, rather than factoring the representation into $(\eta_A, \eta_B, \mathrm{post}_A)$, the model might represent a single augmented belief state over the joint space of world states and components. This will result in a high-dimensional space with a more complex simplex geometry. Another alternative is that the model might represent only the marginal next-token distribution $P(x_{t+1} \mid x_{0:t})$ directly, without explicitly factoring it into per-component belief states. This would be a lower-dimensional representation.

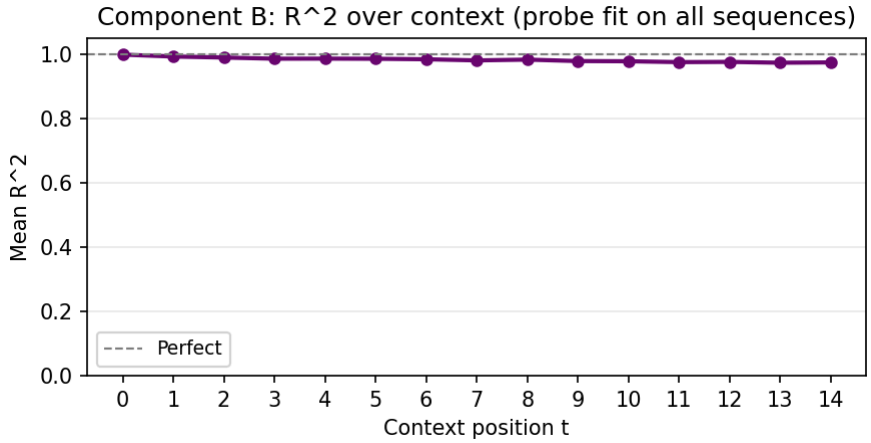
I first test the hypothesis that the model accumulates evidence across context positions to distinguish the latent component. I fit a logistic regression classifier on the final-block residual stream to predict the true underlying component label. Classification accuracy starts at chance at $t = 0$ and increases monotonically, consistent with prediction that the model accumulating evidence about component identity over context.

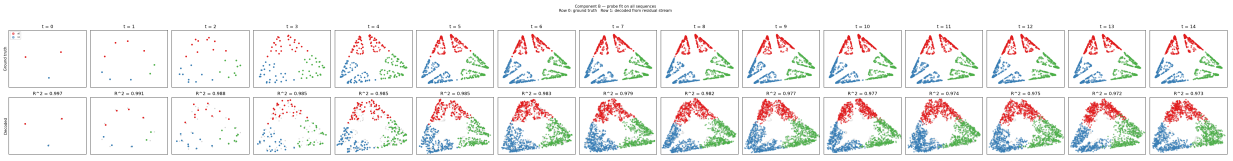


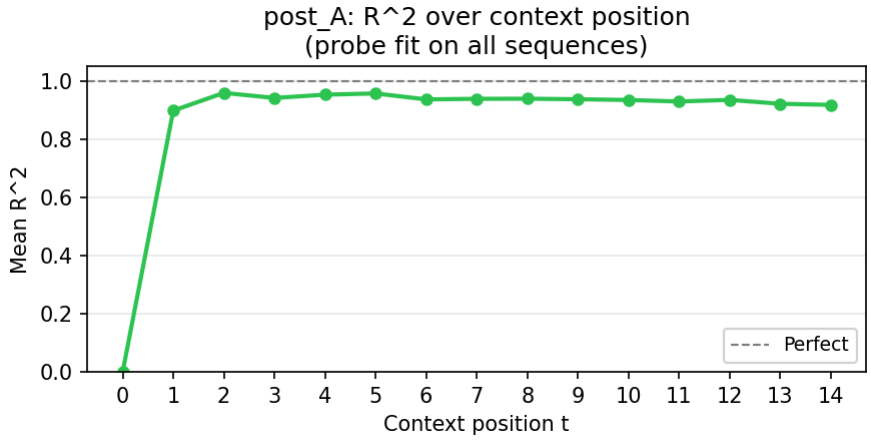
To test whether the model jointly represents $\eta_A(t)$, $\eta_B(t)$, and post_A, I fit OLS linear probes from the final-block residual stream to each belief state. Critically, probes were fit on sequences from both underlying components, meaning for a sequence actually generated by component B, the probe for η_A is decoding the belief state the model would assign *if the sequence came from component A*, and vice versa. For $\eta_A(t)$ and $\eta_B(t)$, both probes achieved $R^2 \approx 1.0$ uniformly across all context positions. This confirms that the residual stream simultaneously and linearly encodes both belief states throughout the entire sequence, regardless of which component generated it. This is not a trivial finding: it implies that for a sequence generated by component A, the model is actively maintaining and updating the belief state it would have under component B, and vice versa. This is consistent with the Bayesian computation above, where both belief states are needed at every step. For post_A, the decoding start at chance at $t = 0$ and increas to $R^2 = 0.9$, consistent with the hypothesis that the initial token is not informative for distinguishing between the two latent components, but the residual stream builds up a linearly decodable representation of component identity as evidence accumulates.



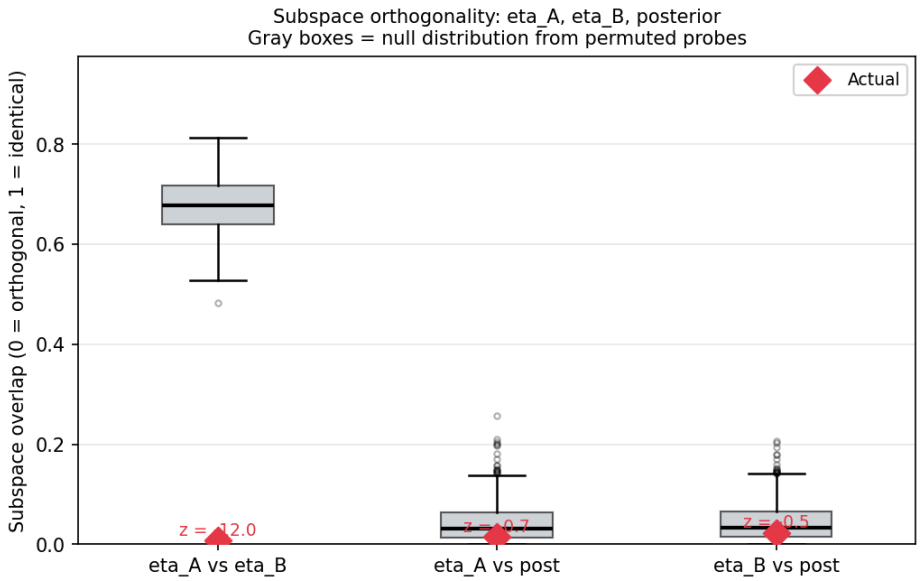




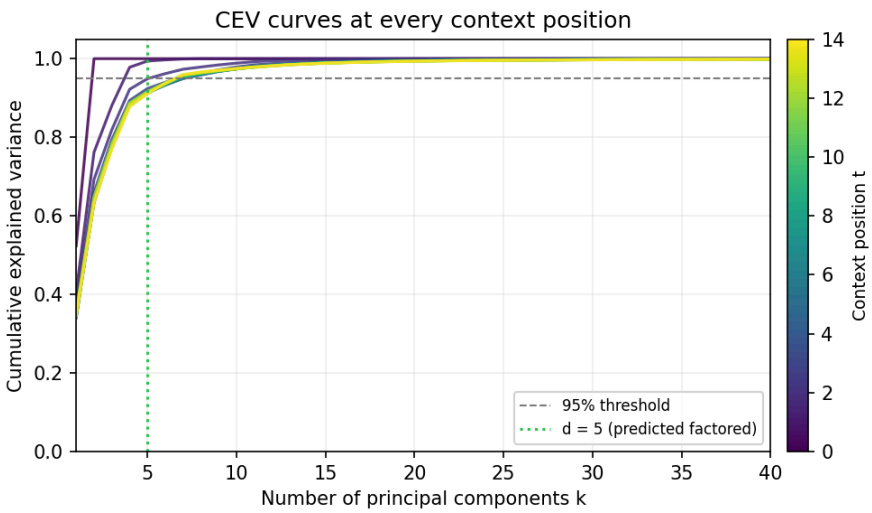




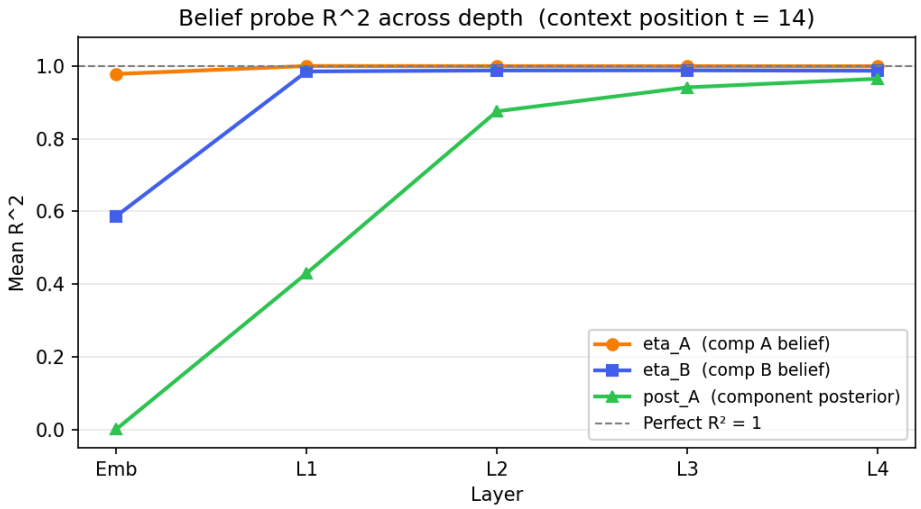
To examine the geometric relationship between these three representations, I extracted the subspaces they occupy in the residual stream. For each probe weight matrix, I obtained an orthonormal basis via SVD, which is a 2D subspace for η_A and η_B respectively and a 1D subspace for post_A. I then measured pairwise subspace overlap using the metric $\frac{1}{d_{\min}}\|Q_1^\top Q_2\|_F^2$, which ranges from 0 to 1 (Shai et al., 2026). To assess significance, I compared against a null distribution obtained by fitting probes to permuted targets. The overlap between the η_A and η_B subspaces was close to 0 compared to a null mean ($z = 12.0$), providing strong evidence that the two belief-state subspaces are nearly orthogonal. The overlap between $\mathrm{post}_A(t)$ and $\eta(t)$ subspaces is smaller than chance, but did not seem to differ significantly. These non-significant results appear surprising, but might not provide strong evidence against orthogonality. The null means are already near the floor, leaving little room for the actual overlap to fall significantly below chance. Whether post_A is genuinely orthogonal to the belief subspaces therefore requires more targeted analysis, which I return to in a later section.



Next, I examined how the dimensionality of this representation evolves with context position. At $t = 0$, the residual stream is nearly rank-1: a single principal component captures essentially all variance, consistent with the prediction that all sequences map to the same point. As context accumulates, the curves flatten progressively, requiring more principal components to reach the 95% threshold. The transition happens rapidly in the first few positions and stabilizes by around $t = 5$ to $t = 7$, after which the curves change only modestly. This confirms the prediction that the residual stream geometry expands from a collapsed point into a structured low-dimensional manifold as the model accumulates evidence.

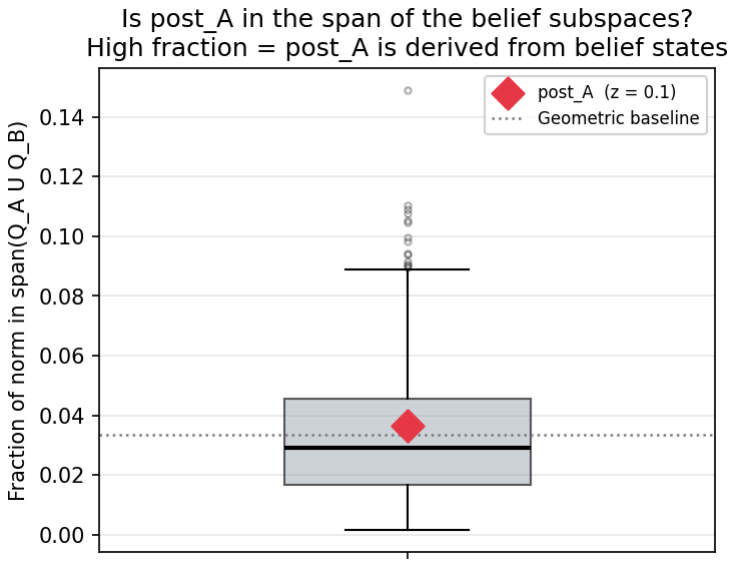


I next examined how these representations build up across transformer depth by evaluating all probes and subspace metrics at the final context position $t = 14$. The η_A curve is near-perfect from the embedding layer onward, which is a direct consequence of the choice of comp A parameters ($\alpha = 0.95$). With such strong self-transitions, the current token almost entirely determines the belief state. The more informative comparison is η_B and post_A. η_B starts at $R^2 = 0.6$ at the embedding and reaches near-perfect decoding by L1, indicating that the transformer resolves the history-dependent belief state of component B within a single layer. post_A is completely undecodable from the embedding and continues climbing gradually. This lag is consistent with the theoretical expectation that inferring component identity requires integrating likelihood ratios across the full token history.

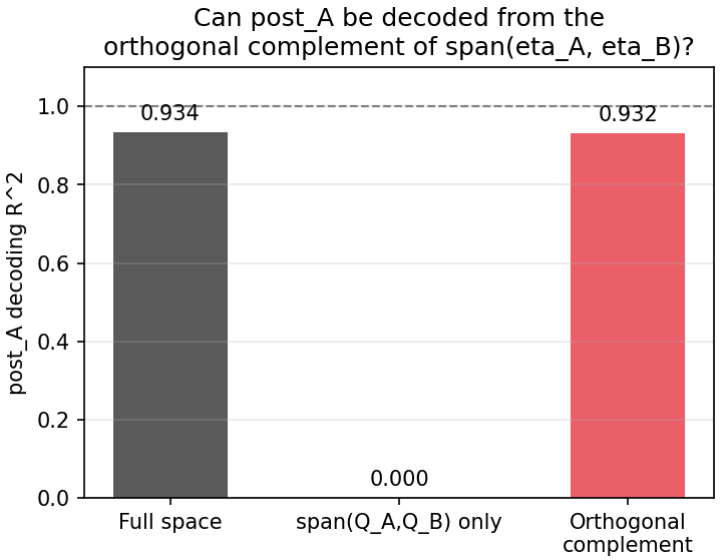


As an additional analysis, I investigated whether the component posterior post_A is really encoded independently of the two belief subspaces, or whether it is a derived quantity computed from η_A and η_B and therefore lies within their span. This question cannot be resolved by the subspace overlap z-scores reported earlier, because the null overlaps between a 2D and a 1D subspace in a high-dimensional space are already close to the floor, leaving little room to detect orthogonality via a z-score. A result near zero is consistent with both genuine orthogonality and with the actual overlap being indistinguishable from noise.

I first measured the fraction of the post_A probe direction's norm recovered by projecting onto $\mathrm{span}(Q_A \cup Q_B)$. The actual fraction was only slightly above the null mean of ($z = 0.1$), suggesting post_A does not preferentially align with the η belief subspaces.



I then performed another analysis: after removing $\mathrm{span}(Q_A \cup Q_B)$ from the activations entirely, a fresh probe for post_A in the residual space achieved $R^2 = 0.932$, nearly identical to the full-space $R^2 = 0.934$. By contrast, decoding post_A from $\mathrm{span}(Q_A \cup Q_B)$ alone yielded $R^2 = 0.000$. This means virtually all post_A information lives outside the η belief subspaces.



I think the most intriguing extention would be to a continuous mixture. Instead of drawing a discrete component label once per sequence, we could draw a continuous parameter $\theta \sim \mathrm{Uniform}(0.05, 0.95)$ and generate the entire sequence from Mess3($\alpha = \theta, x = 0.15$), keeping x fixed so that marginal token frequencies are identical across sequences and the model must rely on temporal correlation structure to infer θ. In this case, the posterior over θ is a distribution over an interval, and the sufficient statistic for Bayes-optimal prediction may trace out a continuous manifold in the residual stream rather than a discrete cluster structure. Since there are effectively infinitely many components, the question of whether and how the model can still maintain parallel belief states for each is super interesting and has clear practical relevance since real-world language is generated by a continuous distribution of latent causes rather than a finite discrete set.

References

Shai, A., Teixeira, L., Oldenziel, A., Marzen, S., & Riechers, P. (2024). Transformers represent belief state geometry in their residual stream. *Advances in Neural Information Processing Systems*, *37*, 75012-75034.

Shai, A., Amdahl-Culleton, L., Christensen, C. L., Bigelow, H. R., Rosas, F. E., Boyd, A. B., ... & Riechers, P. M. (2026). Transformers learn factored representations. *arXiv preprint arXiv:2602.02385*.

Ortega, P. A., Wang, J. X., Rowland, M., Genewein, T., Kurth-Nelson, Z., Pascanu, R., ... & Legg, S. (2019). Meta-learning of sequential strategies. *arXiv preprint arXiv:1905.03030*.

Genewein, T., Wenliang, L. K., Grau-Moya, J., Ruoss, A., Orseau, L., & Hutter, M. (2025). Understanding prompt tuning and in-context learning via meta-learning. *arXiv preprint arXiv:2505.17010*.